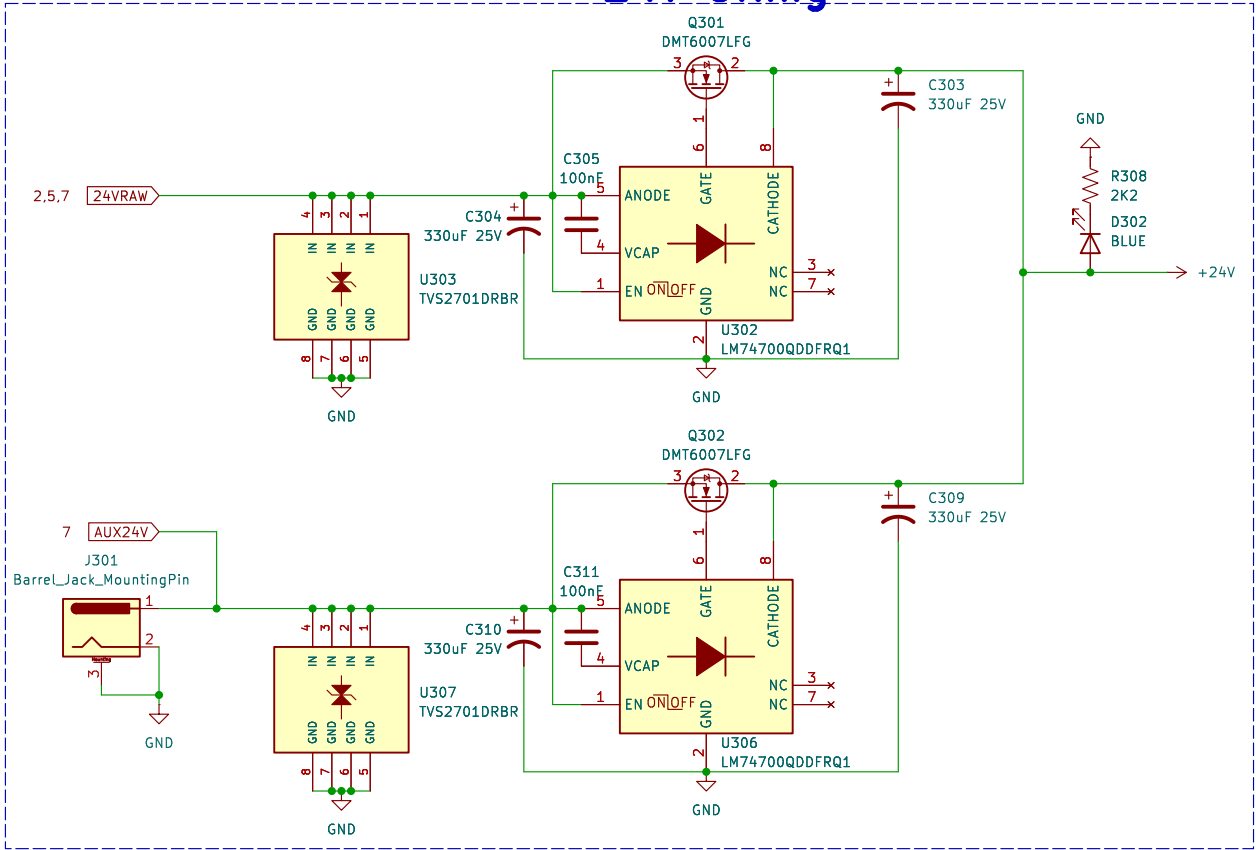
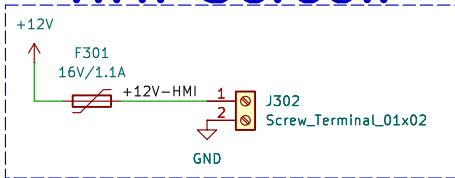


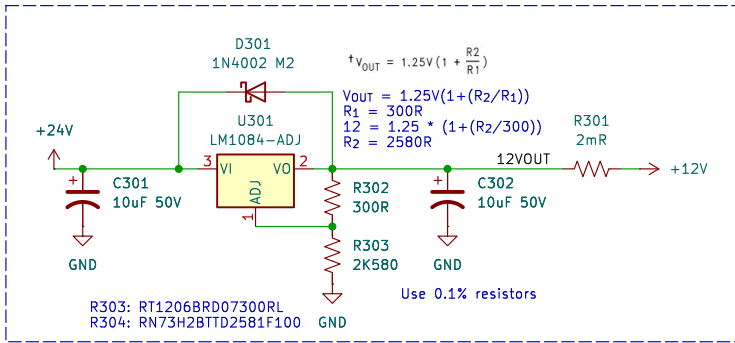
24V ORing



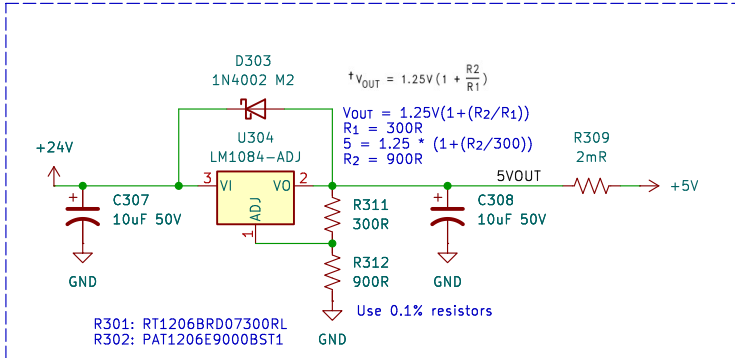
HMI Screen



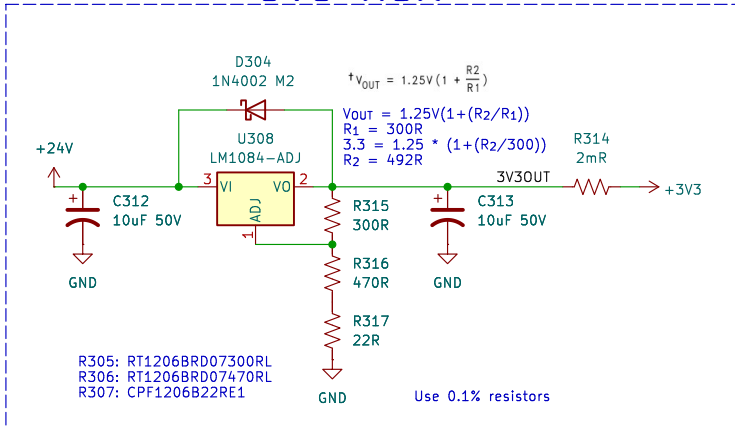
12V Rail



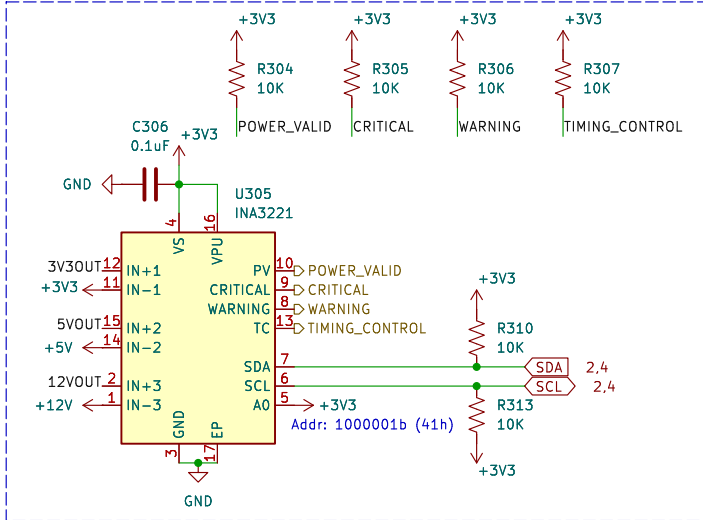
5V Rail



3V3 Rail



Current & Power Monitor



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)

McMillan Enterprises

Sheet: /External Power/
File: external_power.kicad_sch

Title: MEP MAXI 85V-1 REV 2 Interface Board

Size: A3

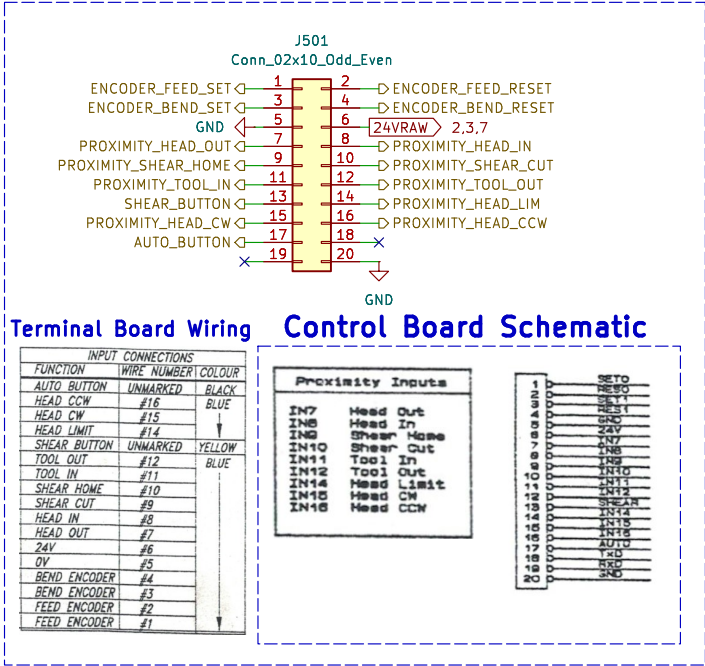
Date: 2025-07-12

Rev: X3

KiCad E.D.A. 9.0.3

Id: 3/9

INPUT CONNECTIONS



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)

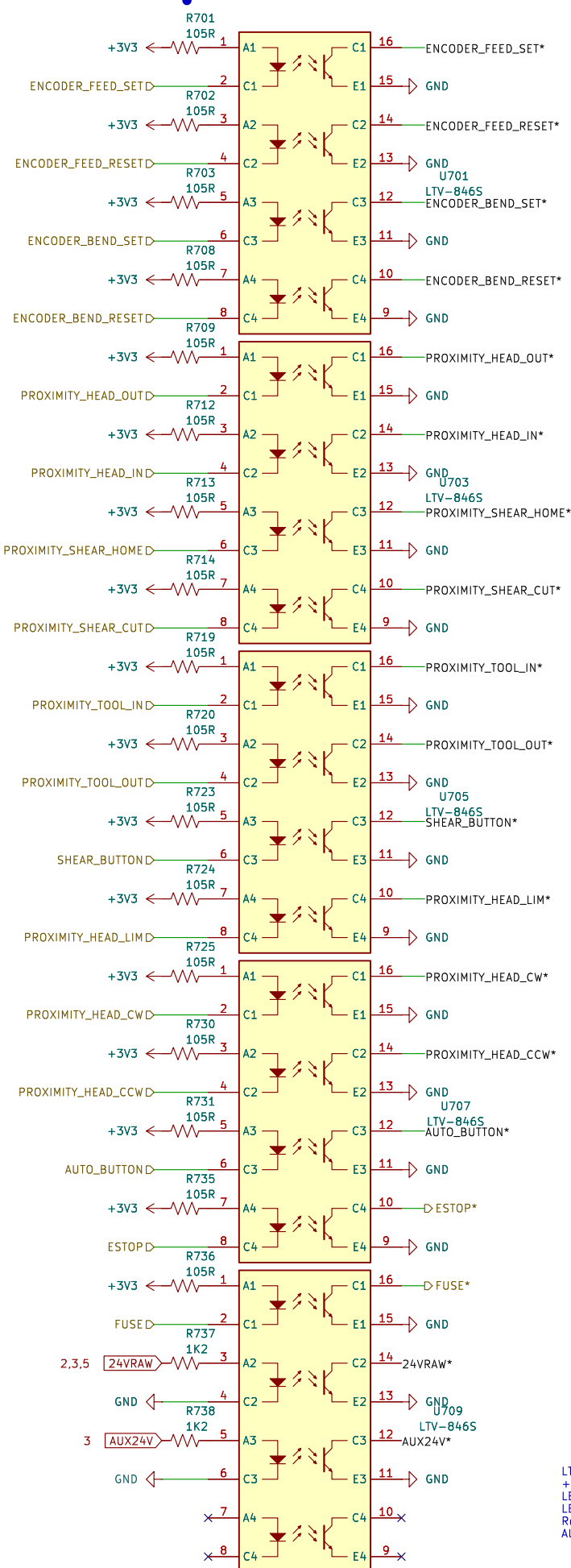
McMillan Enterprises

Sheet: /Input Connections/
File: input_connections.kicad_sch

Title: MEP MAXI 85V-1 REV 2 Interface Board

Size: A3	Date: 2025-07-12	Rev: X3
KiCad E.D.A. 9.0.3		Id: 5/9

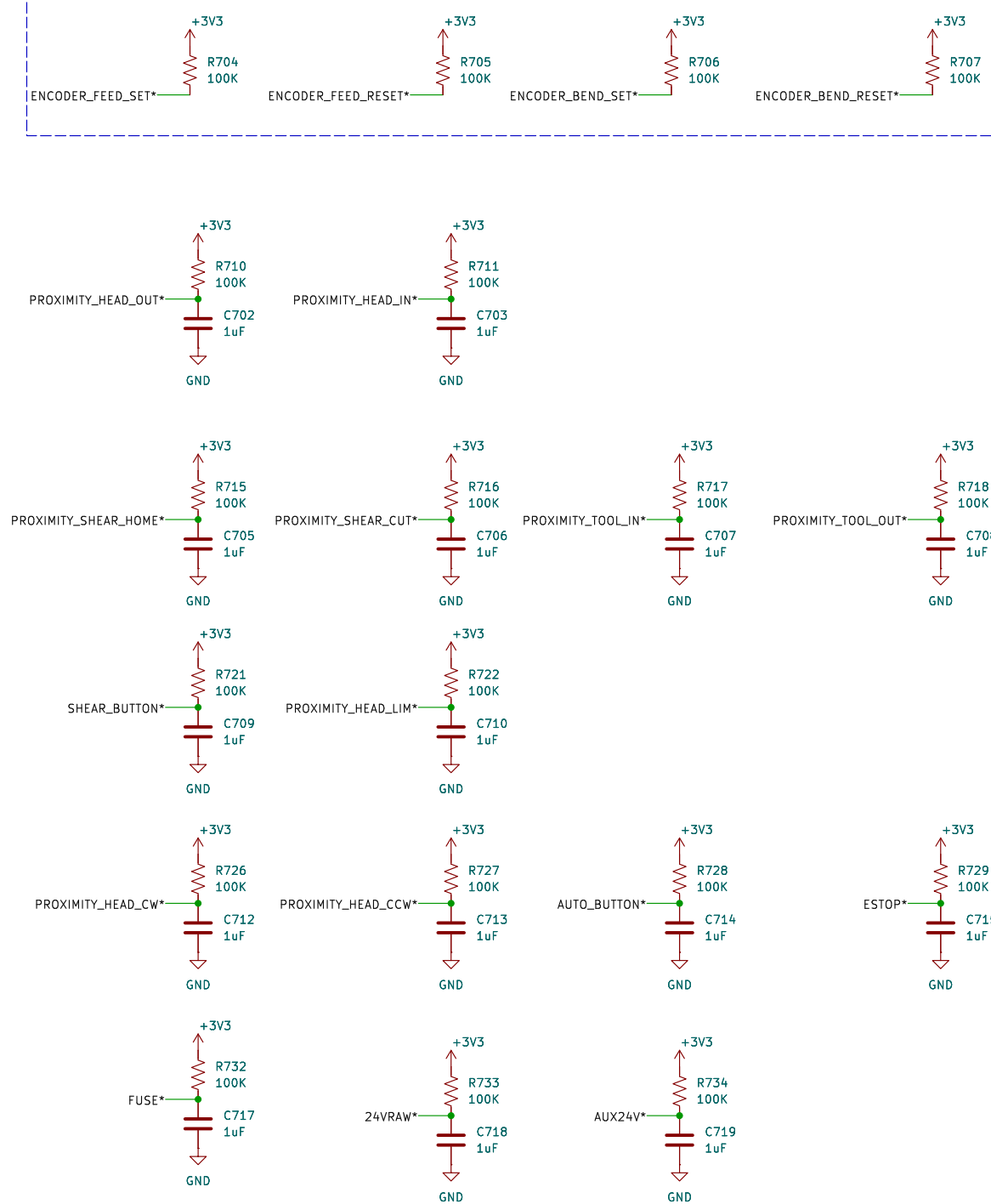
Optoisolators



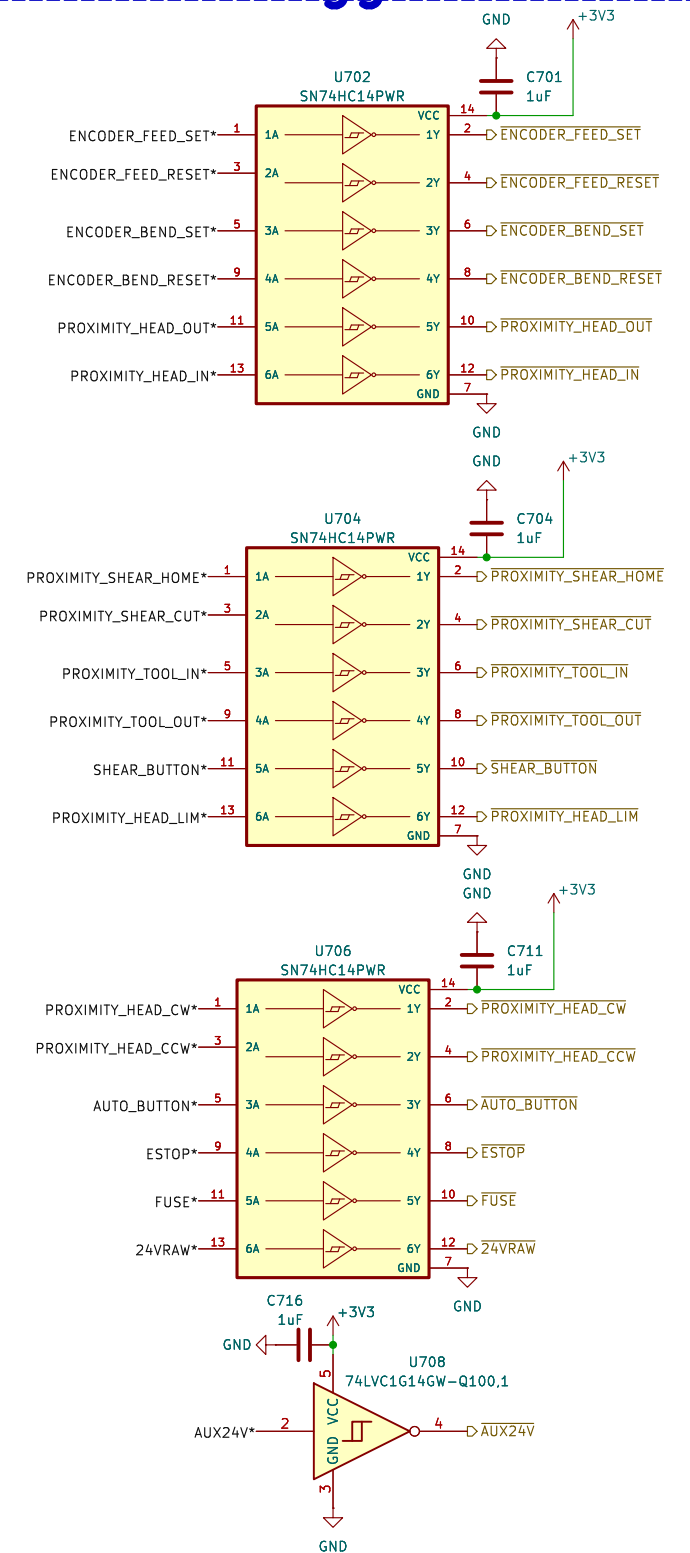
LTV-8465
+V Value: 3.3V
LED Voltage Drop: 1.2V
LED Current: 20mA
Resistor Value: 105R
Alternative Value: 110R

Delay

Encoder signals should not be delayed, otherwise the pulses of the encoder will not register above a certain RPM.

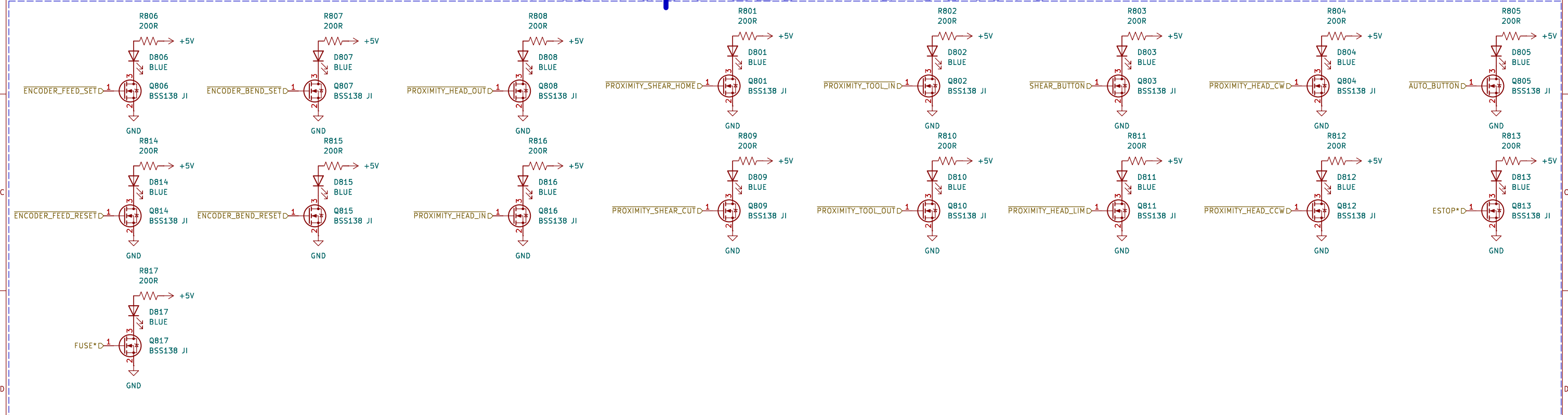


Schmitt Trigger Inverters



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)
McMillan Enterprises
Sheet: /Signal Processing/
File: signal-processing.kicad_sch
Title: MEP MAXI 85V-1 REV 2 Interface Board
Size: A3 Date: 2025-07-12 Rev: X3
KiCad E.D.A. 9.0.3 Id: 7/9

Interrupt LED Indicators



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)

McMillan Enterprises

Sheet: /Interrupt LEDs/
File: interrupt-leds.kicad_sch

Title: MEP MAXI 85V-1 REV 2 Interface Board

Size: A3

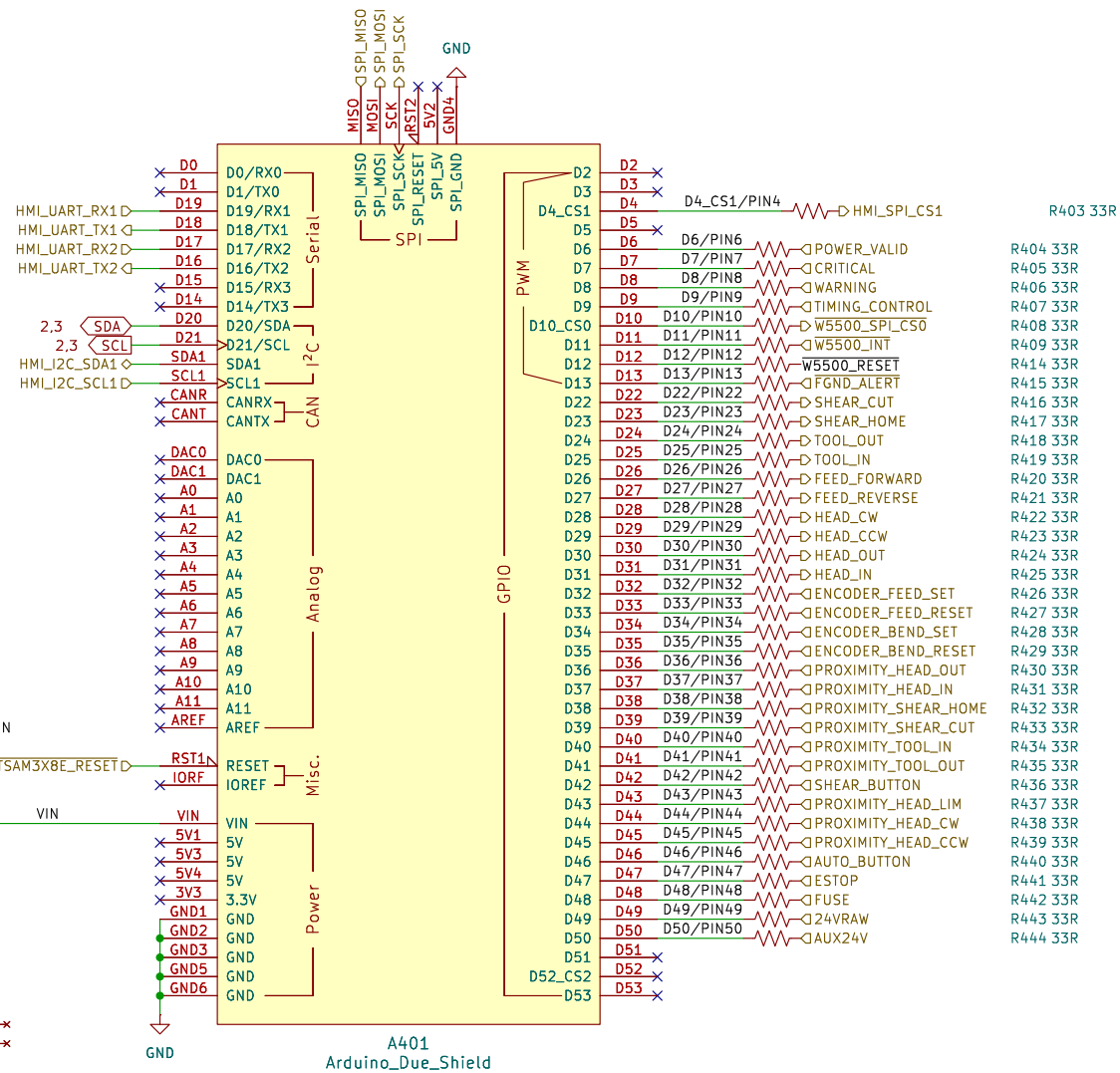
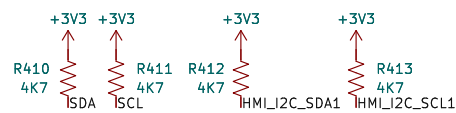
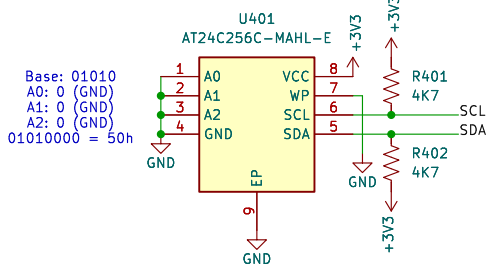
Date: 2025-07-12

Rev: X3

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Id: 8/9

I2C EEPROM

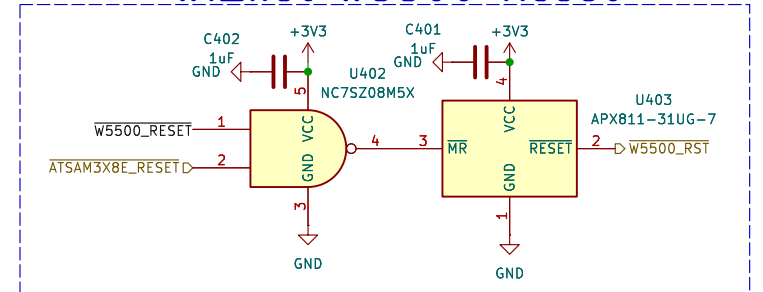


Pin Description

Pin#	I/O	Description
D22/PB26	O	SHEAR_CUT
D23/PA14	O	SHEAR_HOME
D24/PA15	O	TOOL_OUT
D25/PD0	O	TOOL_IN
D26/PD1	O	FEED_FORWARD
D27/PD2	O	FEED_REVERSE
D28/PD3	O	HEAD_CW
D29/PD6	O	HEAD_CCW
D30/PD9	O	HEAD_OUT
D31/PA7	O	HEAD_IN
D32/PD10	I	ENCODER_FEED_SET
D33/PC1	I	ENCODER_FEED_RESET
D34/PC2	I	ENCODER_BEND_SET
D35/PC3	I	ENCODER_BEND_RESET
D36/PC4	I	PROXIMITY_HEAD_OUT
D37/PC5	I	PROXIMITY_HEAD_IN
D38/PC6	I	PROXIMITY_SHEAR_HOME
D39/PC7	I	PROXIMITY_SHEAR_CUT
D40/PC8	I	PROXIMITY_TOOL_IN
D41/PC9	I	PROXIMITY_TOOL_OUT
D42/PA19	I	SHEAR_BUTTON
D43/PA20	I	PROXIMITY_HEAD_LIMIT
D44/PC19	I	PROXIMITY_HEAD_CW
D45/PC18	I	PROXIMITY_HEAD_CCW
D46/PC17	I	AUTO_BUTTON
D47/PC16	I	ESTOP
D48/PC15	I	FUSE

I = Interrupt Input

WIZnet W5500 Reset



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)

McMillan Enterprises

Sheet: /Arduino Due/
File: arduino_due_io.kicad_sch

Title: MEP MAXI 85V-1 REV 2 Interface Board

Size: A3

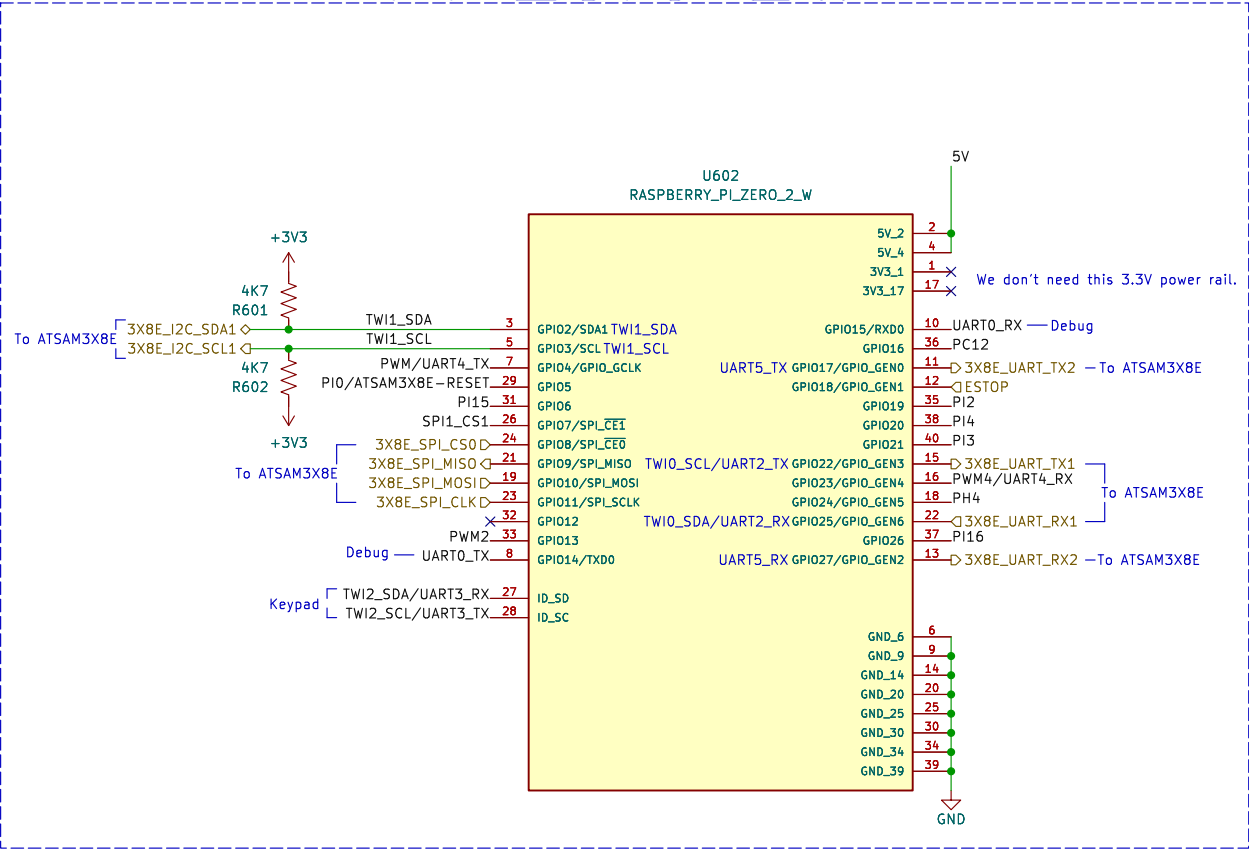
Date: 2025-07-12

Rev: X3

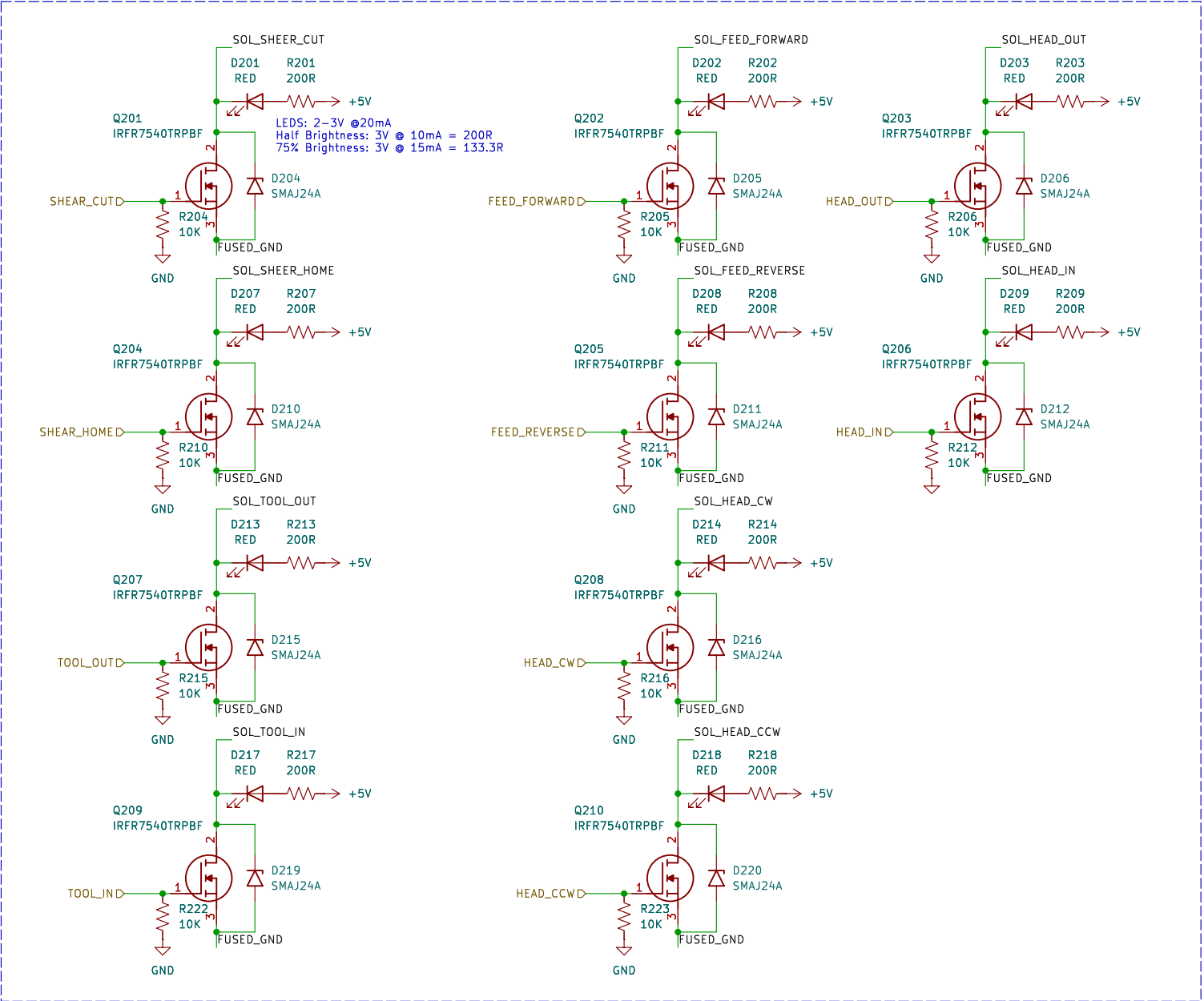
KiCad E.D.A. 9.0.3



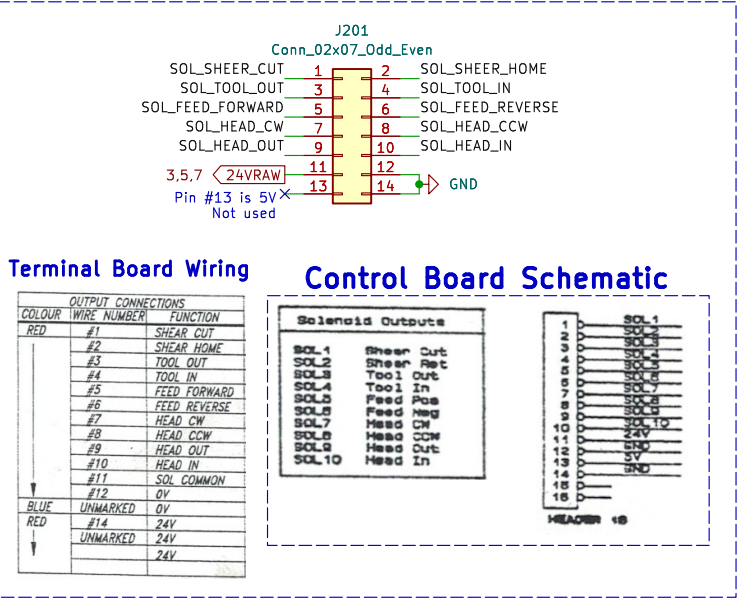
Pi Zero 2W



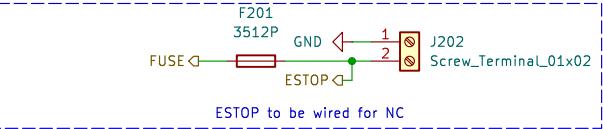
Solenoid MOSFETs



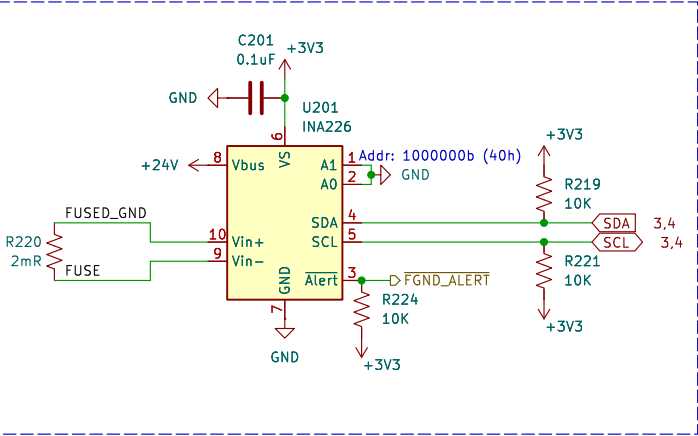
MEP MAXI



ESTOP



Power & Current Monitor



Hardware: Connor McMillan (connor@mcmillan.website)
Firmware: John Gilcreast (johngilcreast@gmail.com)

McMillan Enterprises

Sheet: /Output Connections/
File: output_connections.kicad_sch

Title: MEP MAXI 85V-1 REV 2 Interface Board

Size: A3 Date: 2025-07-12
KiCad E.D.A. 9.0.3

Rev: X3
Id: 2/9